

Lab: Boundary Mapping -- What Makes a System?

Objective

Identify and map the boundaries of systems you encounter every day, using the concept of the **Markov blanket** to distinguish what is "inside" a system from what is "outside."

Prerequisites

- Read the module on Systems and Boundaries
- Bring a notebook or use a drawing app on your device

Part 1: Your Phone as a System

Goal: Map the boundary of your smartphone as a system.

Think of your phone as a system with an inside and an outside. The Markov blanket is the boundary that separates the phone's internal states from the external world.

1. **Internal states:** List five things that are "inside" the phone system (e.g., battery charge, stored photos, app data).
2. **Sensory inputs:** List three ways the phone receives information from the outside (e.g., touchscreen, microphone, camera).
3. **Active outputs:** List three ways the phone acts on the outside world (e.g., speaker, screen display, vibration).
4. **Blanket states:** The sensory inputs and active outputs together form the Markov blanket. Draw a circle with the internal states inside, the external world outside, and the blanket states on the circle's edge.

Write your response here

Part 2: Your School as a System

Goal: Apply the same boundary-mapping to a larger system -- your school.

1. What counts as "internal" to the school system? (Think: students during school hours, teachers, curriculum, building temperature.)
2. What are the school's sensory inputs? (Think: attendance records, test scores, parent emails, fire alarms.)
3. What are the school's active outputs? (Think: report cards, announcements, events, graduates.)
4. Draw or describe the Markov blanket of the school.

Write your response here

Part 3: You as a System

Goal: Apply boundary-mapping to yourself.

1. What is inside your personal system? (Organs, thoughts, memories, feelings.)
2. What are your sensory states? (Eyes, ears, skin, nose, tongue -- plus hunger signals, pain.)
3. What are your active states? (Speaking, moving, facial expressions, typing.)
4. How does your Markov blanket change when you put on headphones? When you are asleep?

Write your response here

Part 4: Comparing Systems

Fill in the comparison table:

Feature	Phone	School	You
Number of sensory channels			
Can it maintain itself without outside help?			
Does it have preferred states (homeostasis)?			
What happens if the boundary breaks?			

Write your response here

Discussion Questions

1. What is the key difference between a system that is "alive" and one that is not, from an Active Inference perspective?
2. Can a system exist without a boundary? Why or why not?
3. Give an example of a system whose boundary is hard to define. What makes it tricky?

Write your response here

Summary

Concept	Definition	Your Example
System	A collection of states separated from the environment by a boundary	
Markov Blanket	The set of sensory and active states forming the boundary	
Internal States	States hidden from the outside world	
Homeostasis	The system's tendency to maintain preferred states	